

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA1678 COURSE NAME: Data Warehousing and Data Mining for Cluster Analysis

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre-processing techniques like Data cleaning, Data integration, Data transformation and data Reduction, for effective data preparation (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 7

Total Marks: 50

Annexure A

SENARIO BASED QUESTIONS

Code of Conduct:

I _Shahana (192524479)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Total (50) Marks	Out of 100
Weightage	20%	20%	25%	20%	15%	100%	
Q1							
Q2							
Q3							
Q4							
Q5							

Signature of the Faculty

Total Marks Awarded :

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

1. Smart Traffic Monitoring

Traffic sensors collect city data.

Sensor_ID	Speed (km/h)	Vehicle_Count	Signal_Time	Area_Type	Congestion
T1	40	200	60	Urban	Low
T2	NaN	180	55	Urban	Medium
T3	50	NaN	70	HIGHWAY	Low
T4	200	1000	200	Highway	High
T5	45	210	NaN	Urban	Medium

Tasks

- A. Missing value treatment
- B. Detect outliers in Speed and Vehicle_Count
- C. Standardize Area_Type
- D. Normalize Signal_Time
- E. Identify mining task

2. Movie Recommendation System

Streaming platform analyzes user behavior.

User ID	Watch Time	Ratings	Reviews	Genre	Preference
U1	10	4	5	Action	High
U2	NaN	3	4	Action	Medium
U3	12	Nan	6	Drama	High
U4	100	10	50	Drama	Low
U5	8	2	Nan	Action	Medium

Tasks

- A. Handle missing values
- B. Detect outliers
- C. Standardize Genre
- D. Discretize Ratings
- E. Mining task

3. Fitness Tracking Dataset

A wearable device collects user data.

User ID	Steps	Calories	Sleep (hrs)	Gender	Fitness_Level
F1	8000	2000	7	Male	Good
F2	NaN	1800	6	Male	Average
F3	9000	NaN	8	Female	Poor
F4	50000	10000	20	Female	Poor
F5	7000	1500	NaN	Female	Average

Tasks

- A. Missing value handling
- B. Detect outliers
- C. Standardize Gender
- D. Normalize Steps
- E. Mining task

4. Weather Prediction System

Weather stations collect environmental data.

Station ID	Temp (°C)	Humidity	Rainfall	Region	Weather
W1	30	70	100	Coastal	Rainy
W2	NaN	65	80	Coastal	Cloudy
W3	32	NaN	90	Inland	Sunny
W4	80	10	500	Inland	Storm
W5	28	75	NaN	Coastal	Rainy

Tasks

- A. Handle missing values
- B. Detect outliers
- C. Standardize Region
- D. Normalize Rainfall
- E. Identify mining functionality

5. Employee Performance Analysis

HR collects performance metrics.

Emp_ID	Work_Hours	Projects	Salary (₹K)	Dept	Rating
E1	40	5	30	IT	High
E2	NaN	4	35	IT	Medium
E3	45	NaN	40	HR	High
E4	20	20	200	HR	Low
E5	38	3	NaN	HR	Medium

Tasks

- A. Missing value imputation
- B. Outlier detection
- C. Standardize Dept
- D. Discretize Salary
- E. Mining task selection

6. Supply Chain Optimization

Company tracks logistics data.

Order_ID	Delivery_Time	Distance	Cost (₹K)	Transport	Status
O1	5	100	10	Truck	On-Time
O2	NaN	120	12	Truck	Delayed
O3	6	NaN	11	SHIP	On-Time
O4	50	1000	200	Ship	Delayed
O5	4	90	NaN	Truck	On-Time

Tasks

- A. Missing value treatment
- B. Detect outliers
- C. Standardize Transport
- D. Normalize Distance
- E. Mining task

7. Banking Fraud Detection

A bank analyzes transaction data.

Txn_ID	Amount (₹K)	Frequency	Balance (₹K)	Mode	Fraud
T1	10	5	50	Online	No
T2	NaN	8	60	Online	No
T3	20	NaN	55	CARD	Yes
T4	500	50	2000	Card	Yes
T5	12	6	NaN	Online	No

Tasks

- A. Data cleaning
- B. Detect abnormal transactions
- C. Standardize Mode
- D. Discretize Amount
- E. Mining task selection

1. Smart Traffic Monitoring

Data:

ID Speed Vehicle_Count Signal_Time Area_Type Congestion

T1	40	200	60	Urban	Low
T2	NaN	180	55	Urban	Medium
T3	50	NaN	70	HIGHWAY	Low
T4	200	1000	200	Highway	High
T5	45	210	NaN	Urban	Medium

A. Missing Value Treatment We use **median imputation** instead of mean because T4 contains extreme values (Speed=200, Vehicle_Count=1000, Signal_Time=200) that would drag the mean far away from what's "typical." Median is robust to such outliers.

- Speed column: {40, 50, 200, 45} → sorted {40,45,50,200} → median = $(45+50)/2 = 47.5$ → fills T2
- Vehicle_Count column: {200,180,1000,210} → sorted {180,200,210,1000} → median = $(200+210)/2 = 205$ → fills T3
- Signal_Time column: {60,55,70,200} → sorted {55,60,70,200} → median = $(60+70)/2 = 65$ → fills T5

B. Outlier Detection (Speed & Vehicle_Count) Using the IQR rule conceptually (Q1, Q3, and bounds = $Q1 - 1.5 \times IQR$ to $Q3 + 1.5 \times IQR$), or simply visual/domain reasoning since n is small:

- Speed values are ~40–50 km/h for T1,T2,T3,T5, but T4=200 km/h — physically unrealistic for urban/highway traffic sensors → **T4 is an outlier**.
- Vehicle_Count values are ~180–210 for others, but T4=1000 → nearly 5x higher → **T4 is an outlier** here too.
- Conclusion: **Row T4 is a global outlier**, possibly a faulty sensor reading or a data entry error (extra zero). It should be flagged/investigated before model training, not silently deleted.

C. Standardize Area_Type Raw values: {Urban, Urban, HIGHWAY, Highway, Urban}
Standardization means enforcing one consistent case/format (usually Title Case or lowercase):

- HIGHWAY → Highway
- Final categories: **{Urban, Highway}** — a clean 2-level categorical column, ready for one-hot/label encoding.

D. Normalize Signal_Time Using **Min-Max Normalization**: $x' = (x - \min) / (\max - \min)$

After imputation, Signal_Time = {60, 55, 70, 200, 65}, min=55, max=200

ID Signal_Time Normalized

T1	60	$(60-55)/145 = 0.034$
T2	55	0.000
T3	70	$(70-55)/145 = 0.103$
T4	200	1.000
T5	65 (imputed)	$(65-55)/145 = 0.069$

E. Identify Mining Task This is a **Classification** problem — the target variable Congestion (Low/Medium/High) is categorical, and we're predicting it from Speed, Vehicle_Count, Signal_Time, and Area_Type. Secondly, the sensor anomaly in T4 makes this also an **Anomaly/Outlier Detection** task (identifying faulty sensors before classification).

2. Movie Recommendation System

Data:

ID Watch Time Ratings Reviews Genre Preference

U1	10	4	5	Action	High
U2	NaN	3	4	Action	Medium
U3	12	NaN	6	Drama	High
U4	100	10	50	Drama	Low
U5	8	2	NaN	Action	Medium

A. Handle Missing Values Median chosen again because U4 is an extreme value that would distort a mean.

- Watch Time: {10,12,100,8} → sorted {8,10,12,100} → median = $(10+12)/2 = 11$ → fills U2
- Ratings: {4,3,10,2} → sorted {2,3,4,10} → median = $(3+4)/2 = 3.5$ → fills U3
- Reviews: {5,4,6,50} → sorted {4,5,6,50} → median = $(5+6)/2 = 5.5$ → fills U5

B. Detect Outliers

- Watch Time: U1,U2,U3,U5 cluster around 8–12 minutes/hours, but U4=100 — massively higher.
- Ratings: normal scale appears to be 1–5 (since U1,U2,U3,U5 are 2–4), but U4=10, which is out of the expected rating scale entirely — likely a data entry error (e.g., meant 10/10 scale mixed with a 5-point scale) or a genuine extreme engagement outlier.
- Reviews: U4=50 vs others' 4–6 — also an outlier.
- **Conclusion: U4 is a clear multi-attribute outlier**, and interestingly its Preference is "Low" despite huge engagement — worth checking if this is mislabeled or represents a bot/binge-watch anomaly.

C. Standardize Genre Raw: {Action, Action, Drama, Drama, Action} — already consistently capitalized, so standardization here just confirms uniform casing and trims whitespace. Final categories: **{Action, Drama}**.

D. Discretize Ratings Convert continuous Ratings into categorical bins (equal-width binning), e.g. on a 1–10 scale:

- **Low:** 1–3
- **Medium:** 4–6
- **High:** 7–10

Applying to imputed data {4, 3, 3.5, 10, 2}:

ID	Rating	Bin
U1	4	Medium
U2	3	Low
U3	3.5 (imputed)	Low
U4	10	High
U5	2	Low

E. Mining Task Primarily **Classification** (predicting Preference: High/Medium/Low from behavioral features) combined with **Recommendation Mining** (collaborative or content-based filtering using Genre + Ratings + Watch Time to recommend similar movies to similar users).

3. Fitness Tracking Dataset

Data:

ID Steps Calories Sleep Gender Fitness_Level

F1	8000	2000	7	Male	Good
F2	NaN	1800	6	Male	Average

ID Steps Calories Sleep Gender Fitness_Level

F3 9000 NaN 8 Female Poor
F4 50000 10000 20 Female Poor
F5 7000 1500 NaN Female Average

A. Missing Value Handling

- Steps: {8000,9000,50000,7000} → sorted {7000,8000,9000,50000} → median = $(8000+9000)/2 = \mathbf{8500}$ → fills F2
- Calories: {2000,1800,10000,1500} → sorted {1500,1800,2000,10000} → median = $(1800+2000)/2 = \mathbf{1900}$ → fills F3
- Sleep: {7,6,8,20} → sorted {6,7,8,20} → median = $(7+8)/2 = \mathbf{7.5}$ → fills F5

B. Detect Outliers

- Steps: F4=50000 vs normal range 7000–9000 → a person doesn't walk 50,000 steps/day realistically (~35km) → outlier/likely device malfunction or data entry error (extra zero).
- Calories: F4=10000 kcal burned in a day is physiologically extreme for a normal person → outlier.
- Sleep: F4=20 hours of sleep is impossible in a day → outlier, confirms F4 is a corrupted/erroneous record across the board.
- Conclusion:** F4 should be flagged and either corrected (check if decimal/unit error, e.g. 5000 read as 50000) or removed if unverifiable — note it's labeled "Poor" fitness despite huge activity, reinforcing it's a data error, not a real extreme athlete.

C. Standardize Gender Raw: {Male, Male, Female, Female, Female} — already uniform casing; standardization ensures consistent labeling. Final: **{Male, Female}**.

D. Normalize Steps Min-Max Normalization: $x' = (x - \min) / (\max - \min)$ After imputation, Steps = {8000, 8500, 9000, 50000, 7000}, min=7000, max=50000

ID	Steps	Normalized
F1	8000	$(8000-7000)/43000 = 0.023$
F2	8500 (imputed)	0.035
F3	9000	0.047
F4	50000	1.000
F5	7000	0.000

E. Mining Task Classification — predicting Fitness_Level (Good/Average/Poor) from Steps, Calories, Sleep, Gender. Also involves **Outlier/Anomaly Detection** as a preprocessing step given F4's implausible values.

4. Weather Prediction System

Data:

ID Temp Humidity Rainfall Region Weather

W1	30	70	100	Coastal	Rainy
W2	NaN	65	80	Coastal	Cloudy
W3	32	NaN	90	Inland	Sunny
W4	80	10	500	Inland	Storm
W5	28	75	NaN	Coastal	Rainy

A. Handle Missing Values

- Temp: {30,32,80,28} → sorted {28,30,32,80} → median = $(30+32)/2 = \mathbf{31}$ → fills W2
- Humidity: {70,65,10,75} → sorted {10,65,70,75} → median = $(65+70)/2 = \mathbf{67.5}$ → fills W3
- Rainfall: {100,80,90,500} → sorted {80,90,100,500} → median = $(90+100)/2 = \mathbf{95}$ → fills W5

B. Detect Outliers

- Temp: W4=80°C is not physically realistic on Earth's surface (normal range ~25–35°C here) → outlier/sensor fault.
- Humidity: W4=10% is unusually low compared to 65–75% elsewhere — combined with 80°C, strongly suggests sensor malfunction rather than a real weather event.
- Rainfall: W4=500mm vs 80–100mm elsewhere → extreme, consistent with "Storm" label, but still statistically an outlier that needs verification (could be a real storm reading, unlike Temp/Humidity which look erroneous).
- **Conclusion:** W4 is an outlier across all three numeric fields; Rainfall=500 might be genuine (storms do cause heavy rain) but Temp=80°C is almost certainly a sensor error — this row needs manual review rather than blind imputation/removal.

C. Standardize Region Raw: {Coastal, Coastal, Inland, Inland, Coastal} — already uniform. Final categories: **{Coastal, Inland}**.

D. Normalize Rainfall Min-Max: $x' = (x - \min) / (\max - \min)$ After imputation, Rainfall = {100, 80, 90, 500, 95}, min=80, max=500

ID	Rainfall	Normalized
W1	100	$(100-80)/420 = 0.048$
W2	80	0.000
W3	90	0.024
W4	500	1.000
W5	95 (imputed)	0.036

E. Identify Mining Functionality Classification — predicting Weather category (Rainy/Cloudy/Sunny/Storm) from Temp, Humidity, Rainfall, Region. Also **Anomaly Detection** for identifying faulty sensor stations (like W4).

5. Employee Performance Analysis

Data:

ID	Work_Hours	Projects	Salary(₹K)	Dept	Rating
E1	40	5	30	IT	High
E2	NaN	4	35	IT	Medium
E3	45	NaN	40	HR	High
E4	20	20	200	HR	Low
E5	38	3	NaN	HR	Medium

A. Missing Value Imputation

- Work_Hours: {40,45,20,38} → sorted {20,38,40,45} → median = $(38+40)/2 = 39$ → fills E2
- Projects: {5,4,20,3} → sorted {3,4,5,20} → median = $(4+5)/2 = 4.5$ → fills E3
- Salary: {30,35,40,200} → sorted {30,35,40,200} → median = $(35+40)/2 = 37.5$ → fills E5

B. Outlier Detection

- Work_Hours: E4=20 hrs is unusually low compared to 38–45 for others → possible part-time employee or data issue.
- Projects: E4=20 projects vs 3–5 for others → extremely high, inconsistent with only 20 work hours → suspicious/inconsistent record.
- Salary: E4=₹200K vs ₹30–40K for others → far above the rest → outlier.
- **Conclusion:** E4 is internally inconsistent (low hours + high projects + high salary + "Low" rating) — this strongly suggests either a data entry error or a senior/different-role employee (e.g., a manager or contractor) mixed into the dataset who shouldn't be compared directly with regular staff.

C. Standardize Dept Raw: {IT, IT, HR, HR, HR} — already consistent. Final categories: **{IT, HR}**.

D. Discretize Salary Equal-width/domain-based binning:

- **Low:** < ₹35K
- **Medium:** ₹35K–₹50K
- **High:** > ₹50K

ID	Salary	Bin
E1	30	Low
E2	35	Medium
E3	40	Medium
E4	200	High (extreme)
E5	37.5 (imputed)	Medium

E. Mining Task Selection Classification — predicting employee Rating (High/Medium/Low) from Work_Hours, Projects, Salary, Dept. E4's inconsistency also motivates an **Outlier/Anomaly Detection** step before training the classifier, to avoid the model learning from a mislabeled/unusual record.

6. Supply Chain Optimization

Data:

ID	Delivery_Time	Distance	Cost(₹K)	Transport	Status
O1	5	100	10	Truck	On-Time
O2	NaN	120	12	Truck	Delayed
O3	6	NaN	11	SHIP	On-Time
O4	50	1000	200	Ship	Delayed
O5	4	90	NaN	Truck	On-Time

A. Missing Value Treatment

- Delivery_Time: {5,6,50,4} → sorted {4,5,6,50} → median = (5+6)/2 = **5.5** → fills O2
- Distance: {100,120,1000,90} → sorted {90,100,120,1000} → median = (100+120)/2 = **110** → fills O3
- Cost: {10,12,11,200} → sorted {10,11,12,200} → median = (11+12)/2 = **11.5** → fills O5

B. Detect Outliers

- Delivery_Time: O4=50 days vs 4–6 days for others → an order taking 50 days is extreme, likely representing a long-haul/international shipment or a logging error.
- Distance: O4=1000 vs 90–120 for others → consistent with a much longer route, supporting that O4 may be a genuinely different type of shipment (e.g., international vs local).
- Cost: O4=₹200K vs ₹10–12K for others → proportionally scales with the distance, so this may not be a pure "error" but rather a different shipment category that should possibly be modeled separately.
- **Conclusion:** O4 is an outlier, but unlike some other datasets, its extremity is internally *consistent* (long distance → long time → high cost), suggesting it may represent a legitimate different shipment class rather than corrupted data. Still statistically flagged as an outlier for the given dataset's scale.

C. Standardize Transport Raw: {Truck, Truck, SHIP, Ship, Truck}

- SHIP → Ship
- Final categories: **{Truck, Ship}**

D. Normalize Distance Min-Max: $x' = (x - \min) / (\max - \min)$ After imputation, Distance = {100, 120, 110, 1000, 90}, min=90, max=1000

ID	Distance	Normalized
O1	100	(100-90)/910 = 0.011
O2	120	0.033

ID	Distance	Normalized
O3	110 (imputed)	0.022
O4	1000	1.000
O5	90	0.000

E. Mining Task Classification — predicting delivery Status (On-Time/Delayed) from Delivery_Time, Distance, Cost, Transport. This can also be framed as **Regression** if predicting continuous Delivery_Time itself, or **Clustering** if segmenting shipment types (local vs long-haul, as O4 suggests).

7. Banking Fraud Detection

Data:

ID	Amount(₹K)	Frequency	Balance(₹K)	Mode	Fraud
T1	10	5	50	Online	No
T2	NaN	8	60	Online	No
T3	20	NaN	55	CARD	Yes
T4	500	50	2000	Card	Yes
T5	12	6	NaN	Online	No

A. Data Cleaning

- Amount: {10,20,500,12} → sorted {10,12,20,500} → median = $(12+20)/2 = 16$ → fills T2
- Frequency: {5,8,50,6} → sorted {5,6,8,50} → median = $(6+8)/2 = 7$ → fills T3
- Balance: {50,60,55,2000} → sorted {50,55,60,2000} → median = $(55+60)/2 = 57.5$ → fills T5

B. Detect Abnormal Transactions

- Amount: T4=₹500K vs ₹10–20K for others → 25–50x higher → highly abnormal.
- Frequency: T4=50 transactions vs 5–8 for others → unusually high frequency, a classic fraud signature (rapid repeated transactions).
- Balance: T4=₹2000K vs ₹50–60K for others → also far above normal.
- **Conclusion:** T4 is a clear multi-dimensional outlier **and** it is correctly labeled Fraud="Yes" — this is a good sign the outlier detection aligns with the ground truth label, validating that unusually high Amount + Frequency + Balance is a strong fraud indicator. T3 is also labeled Fraud="Yes" despite not being a strong numeric outlier, suggesting fraud isn't purely amount-driven — pattern/behavioral features may matter more than raw magnitude.

C. Standardize Mode Raw: {Online, Online, CARD, Card, Online}

- CARD → Card
- Final categories: **{Online, Card}**

D. Discretize Amount Equal-width/domain bins:

- **Low:** < ₹50K
- **Medium:** ₹50K–₹200K
- **High:** > ₹200K

ID	Amount	Bin
T1	10	Low
T2	16 (imputed)	Low
T3	20	Low
T4	500	High
T5	12	Low

E. Mining Task Selection Classification (predicting Fraud: Yes/No — this is fundamentally a supervised binary classification problem) combined with **Anomaly/Outlier Detection**, since fraud detection in practice often uses anomaly detection techniques (e.g., Isolation Forest, One-Class SVM) precisely because fraud cases are rare and look statistically unusual, as seen with T4.